sPHENIX Internal Note

Double-Interaction MC Efficiency Study

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

sPHENIX Collaboration

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Abstract

This note documents the investigation into the large reconstruction efficiency drop observed in double-interaction Monte Carlo samples compared to single-interaction samples for the sPHENIX isolated photon cross-section measurement (PPG12). The overall photon reconstruction efficiency falls from ∼94% in single-interaction MC to ∼31% in double-interaction MC. We demonstrate that this drop is predominantly a truth-matching artifact caused by vertex displacement, not a genuine physics inefficiency. The signal clusters remain intact in the calorimeter; the matching failure arises because the reconstructed vertex shifts the cluster η beyond the $\Delta R < 0.1$ matching threshold. BDT shower-shape identification is found to be pileup-robust, and true isolation degradation is at the sub-percent level. These findings validate the use of single-interaction MC for efficiency corrections with pileup effects treated as a small systematic uncertainty.

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1 Introduction

The sPHENIX experiment measures the isolated photon cross-section in p+p collisions at $\sqrt{s} = 200$ GeV using the electromagnetic calorimeter (EMCal). The analysis selects isolated photon candidates in the barrel region $|\eta| < 0.7$ and applies an XGBoost-based BDT for shower-shape discrimination between direct photons and hadronic backgrounds.

At RHIC, multiple p+p collisions can occur within a single bunch crossing (pileup or double interaction). When two collisions overlap, the reconstructed vertex is the luminosity-weighted average of the two true vertices. This shifts the apparent vertex position and distorts the reconstructed cluster kinematics (η, E_T) and shower-shape variables.

Full GEANT4 double-interaction MC samples show that the cluster-weighted pileup fractions are significant:

- **0 mrad crossing angle:** 22.4% of clusters come from double-interaction events (18.7% event-level fraction).
- **1.5 mrad crossing angle:** 13.3% of clusters come from double-interaction events (7.2% event-level fraction).

The headline observation motivating this study is a dramatic efficiency drop: the overall photon reconstruction efficiency falls from ∼94% in single-interaction MC to ∼31% in double-interaction MC. Understanding whether this reflects a real physics effect or an analysis artifact is critical for the cross-section measurement.

2 Method

Three complementary investigation macros were developed, each processing 100,000 events from the `photon10` (single-interaction) and `photon10_double` (double-interaction) MC samples. The `photon10` sample covers the truth p_T range 14–30 GeV.

Truth photons are selected with the following criteria:

- Particle ID: `pid` = 22 (photon)
- Photon class: `photonclass` < 3 (direct photons)
- Pseudorapidity: $|\eta_{\text{truth}}| < 0.7$
- Truth isolation: $E_T^{\text{iso,truth}} < 4$ GeV (cone $R = 0.3$)

Truth-to-reco matching uses the standard criterion:

$$\texttt{trkID}\ \text{match} \quad \text{AND} \quad \Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} < 0.1 \tag{1}$$

where the track ID (`trkID`) links the truth particle to the reconstructed cluster, and the ΔR cut ensures spatial consistency.

3 Truth Photon Content

Before examining reconstruction, we verify that the truth-level photon content is essentially identical between single and double-interaction samples. Table 1 summarizes the key quantities.

Table 1: Truth photon content comparison between single and double-interaction MC.

Quantity	**Single**	**Double**
Truth photons per event	0.493	0.495
Fraction with ≥ 2 photons	0.22%	0.21%
Mean truth isolation E_T	0.33 GeV	0.39 GeV

The photon multiplicity and p_T spectra are statistically identical. The small increase in truth isolation energy (0.33 to 0.39 GeV) in double-interaction events is far below the 4 GeV isolation cut and has negligible impact.

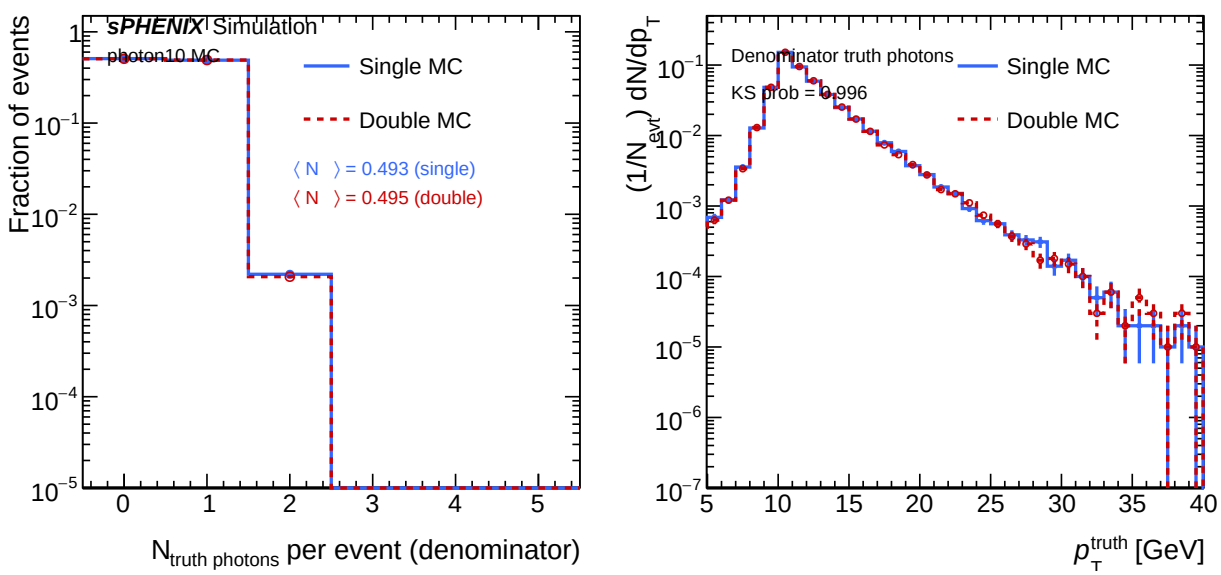

Figure 1: Truth photon multiplicity and p_T distributions for single-interaction (blue) and double-interaction (red) MC samples. The distributions are statistically consistent, confirming that the truth-level photon content is not affected by pileup.

4 Vertex Shift Mechanism

The key to understanding the efficiency drop lies in the vertex displacement. In double-interaction events, the reconstructed vertex is the average of two collision vertices:

$$z_{\text{vtx}}^{\text{reco}} = \frac{z_{\text{vtx},1} + z_{\text{vtx},2}}{2} \tag{2}$$

This introduces a shift $\Delta z = z_{\text{vtx}}^{\text{reco}} - z_{\text{vtx}}^{\text{true}}$ that propagates to the reconstructed cluster pseudorapidity. For a cluster at the EMCal barrel radius $R_{\text{EMCal}} = 93.5$ cm:

$$\Delta\eta \approx -\frac{\Delta z}{R_{\text{EMCal}} \cdot \cosh\eta} \tag{3}$$

The critical threshold for the $\Delta R < 0.1$ matching criterion is:

$$|\Delta z|_{\text{crit}} = 0.1 \times R_{\text{EMCal}} \times \cosh\eta = 9.35 \text{ cm} \quad (\text{at } \eta = 0) \tag{4}$$

When $|\Delta z|$ exceeds this threshold, the cluster η shifts by more than 0.1, and the truth-reco matching fails even though the cluster is physically present and well-reconstructed.

Table 2: Vertex distribution properties for single and double-interaction MC.

Property	**Single**	**Double**
Vertex z RMS	7.6 cm	34.5 cm
Fraction with $|\Delta z| > 9.35$ cm	—	65%

The double-interaction vertex RMS of 34.5 cm is far larger than the single-interaction value of 7.6 cm, and 65% of double-interaction events exceed the critical $|\Delta z|$ threshold. This immediately explains why the bulk of the efficiency loss occurs.

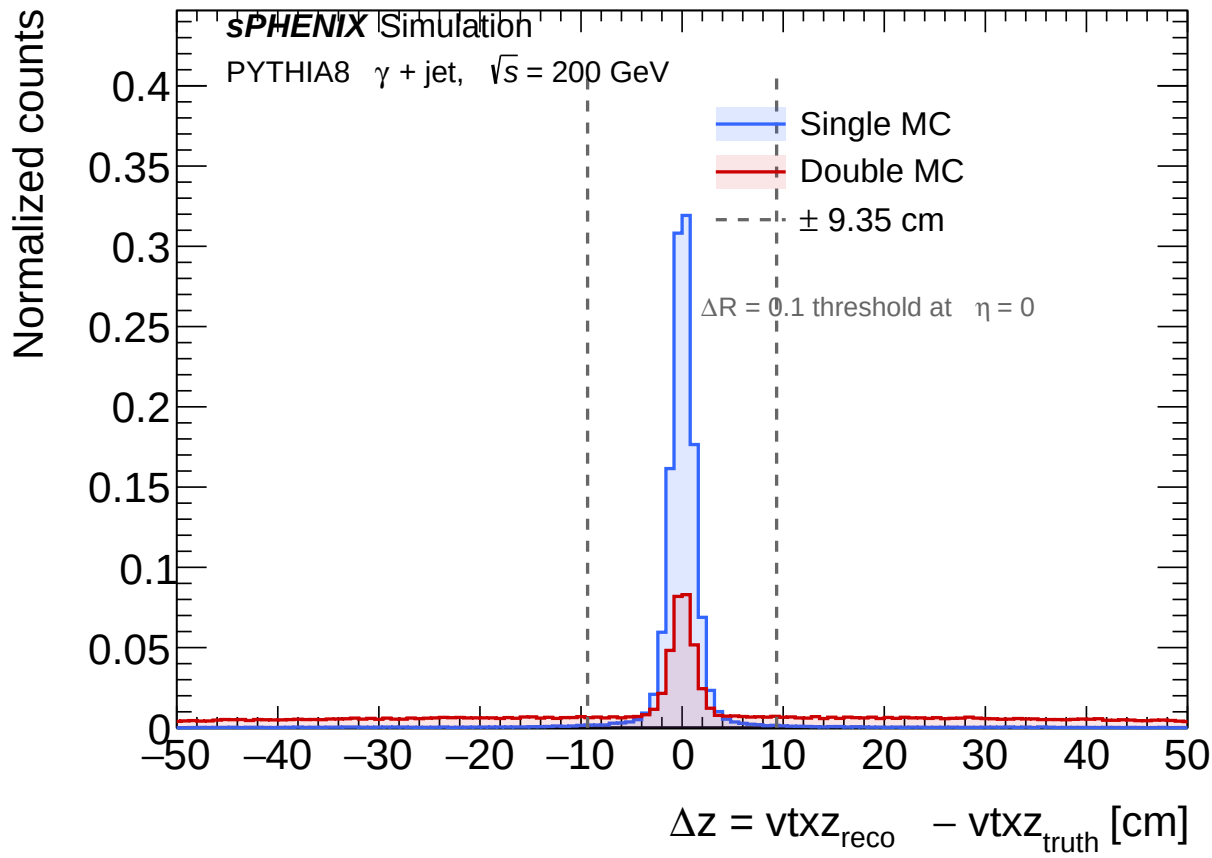

Figure 2: Vertex shift (Δz) distributions for single and double-interaction MC. Vertical lines indicate the critical threshold $|\Delta z| = 9.35$ cm beyond which the $\Delta R < 0.1$ matching criterion fails at $\eta = 0$. Nearly two-thirds of double-interaction events exceed this threshold.

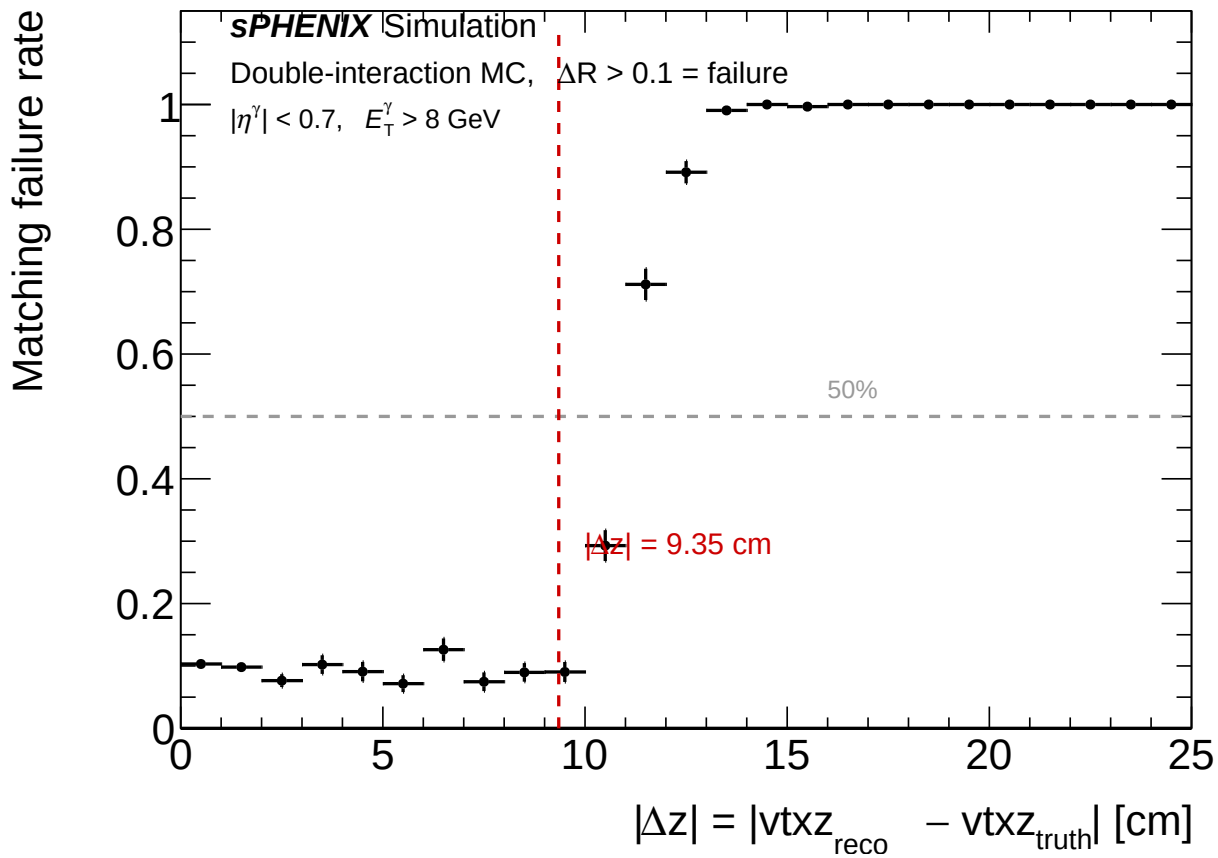

Figure 3: Truth-matching failure rate as a function of $|\Delta z|$. The failure rate exhibits a sharp step at $|\Delta z| \approx 9.35$ cm, confirming the geometric origin of the matching failure.

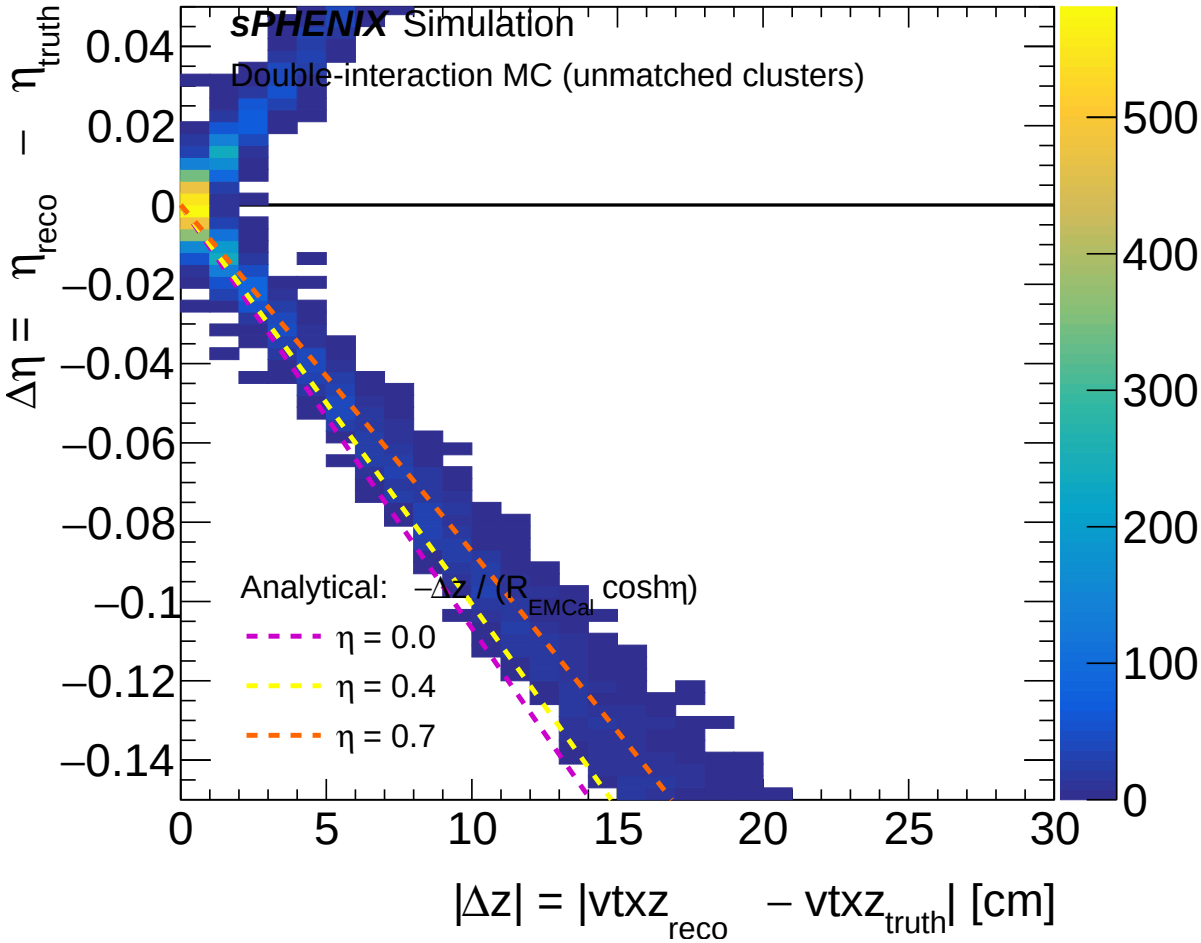

Figure 4: Two-dimensional correlation between $\Delta\eta$ (truth–reco) and Δz (vertex shift). The linear correlation follows the analytic prediction of Eq. (3), providing the "smoking gun" evidence that the matching failure is a geometric artifact of vertex displacement.

5 Decomposition of Efficiency Loss

To quantify the contributions to the efficiency loss, we categorize unmatched truth photons by the reason for the matching failure:

1. **ΔR failure (92%):** A cluster with the correct `trkID` exists in the event, but $\Delta R > 0.1$ due to vertex shift. The signal cluster is physically present and intact.

2. **Missing cluster (8%):** No cluster with the matching `trkID` is found in the event. The cluster was either not reconstructed or merged with another cluster.

3. **Extra photon (∼0%):** Negligible contribution from additional truth photons created by the second interaction.

Table 3: Matching failure rate as a function of $|\Delta z|$ in double-interaction MC.

$\|\Delta z\|$ **range**	**Failure rate**
< 5 cm	$\sim$10%
5–15 cm	$\sim$41%*
15–30 cm	$\sim$100%
> 30 cm	100%

*This range spans the sharp transition at $|\Delta z| = 9.35$ cm; the rate is low at 5 cm and approaches 100% at 14 cm.

The sharp transition around the critical threshold and the dominance of ΔR failures confirm that the efficiency drop is overwhelmingly a matching artifact.

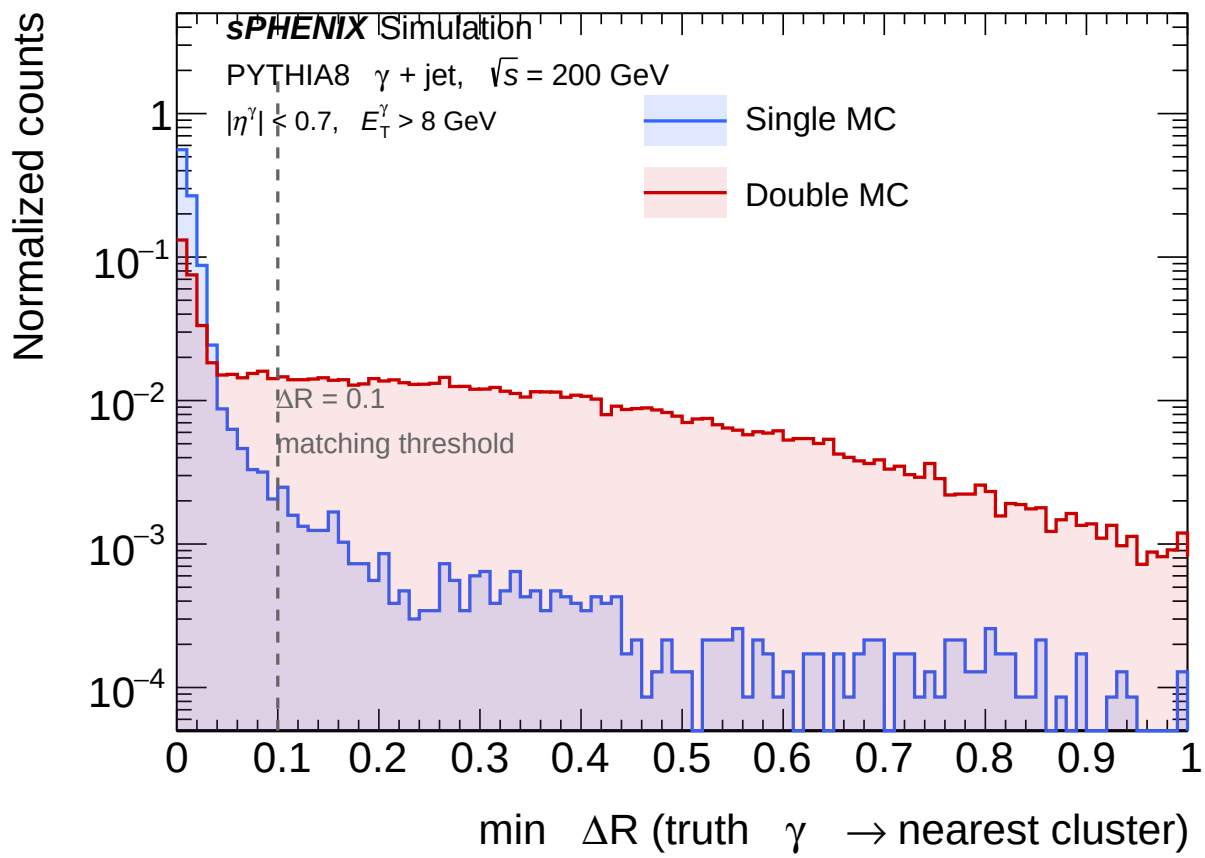

Figure 5: Minimum ΔR between truth photon and the `trkID`-matched reconstructed cluster for single (blue) and double (red) interaction MC. Single-interaction events cluster tightly near $\Delta R = 0$, while double-interaction events show a broad tail extending well beyond the matching threshold of 0.1.

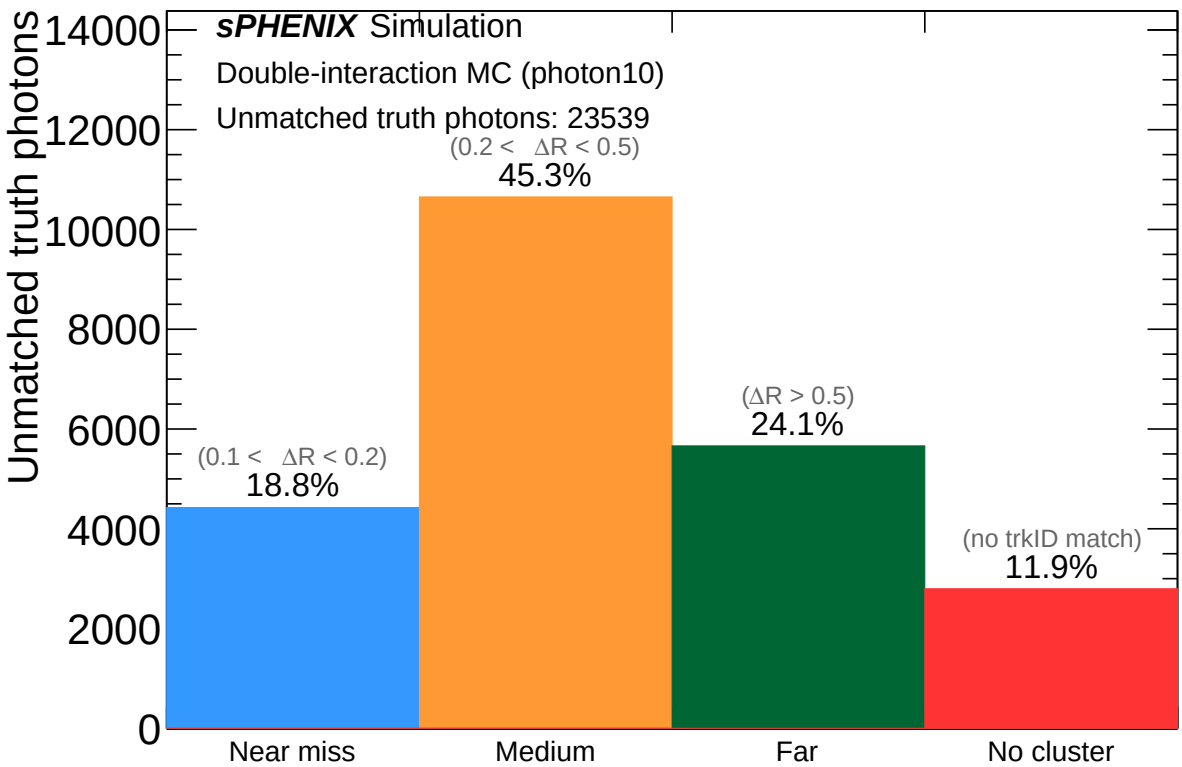

Figure 6: Breakdown of unmatched truth photons in double-interaction MC by failure category. The dominant contribution (92%) is ΔR failure where the cluster exists but the vertex shift moves it beyond the matching threshold. Missing clusters account for 8%, while extra photons from the second interaction are negligible.

6 Per-Efficiency Breakdown

We examine each stage of the photon selection efficiency separately to identify which cuts are genuinely affected by pileup. Table 4 shows the efficiency ranges across p_T bins for each selection stage.

Table 4: Per-stage efficiency ranges across p_T bins for single and double-interaction MC. Ranges are quoted for the `photon10` truth p_T range (14–30 GeV, bins 3+). The first two p_T bins fall below the `photon10` truth threshold and have significantly lower efficiencies due to poor statistics and edge effects (e.g., reco efficiency ∼88%, BDT ∼45–68% for single-interaction MC). The dominant loss occurs at the reconstruction (matching) stage.

Efficiency stage	**Single**	**Double**
Reconstruction (matching)	93–94%	27–32%
BDT shower-shape	81–92%	81–92%
Isolation	61–74%	58–71%
All combined	52–57%	15–19%

Key observations:

- **Reconstruction efficiency** shows the dramatic drop (94% → 31%), driven entirely by the ΔR matching artifact discussed in Sections 4–5.
- **BDT shower-shape efficiency** is identical between single and double-interaction MC (81–92% in the 14–30 GeV range), demonstrating that the BDT is **pileup-robust**. Clusters that pass the matching cut have shower shapes unaffected by the second interaction. (The first two p_T bins, below the `photon10` truth threshold, show significantly lower BDT efficiency due to poor statistics.)
- **Isolation efficiency** shows a modest degradation of ∼4% (absolute) in double-interaction events, reflecting the small amount of additional energy deposited in the isolation cone by the second collision.
- **Combined efficiency** drops from 52–57% to 15–19%, but this is dominated by the reconstruction matching artifact rather than genuine physics inefficiency.

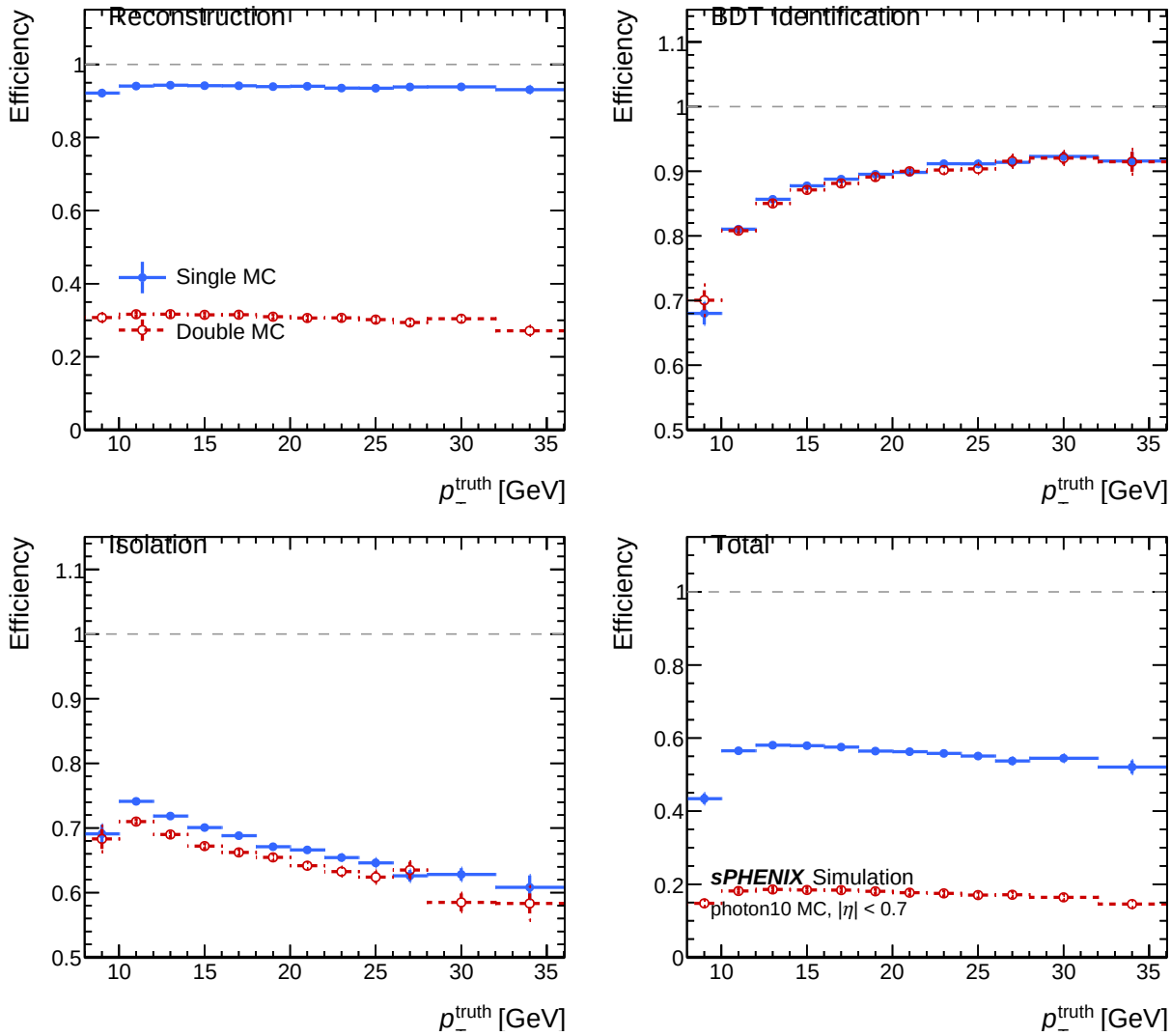

Figure 7: Per-stage efficiency comparison between single-interaction (blue) and double-interaction (red) MC as a function of p_T. The four panels show reconstruction, BDT, isolation, and combined efficiencies. The reconstruction panel exhibits the large pileup-induced drop, while BDT efficiency is unaffected and isolation shows only a small (∼4%) degradation.

7 TEfficiency Bug Fix

During this investigation, a bug was identified in the efficiency calculation code related to ROOT's `TEfficiency` class and weighted events.

The original implementation used:

```
eff->SetWeight(weight);
eff->Fill(passed, x);
```

This approach stores *unweighted* event counts internally, because `SetWeight` followed by `Fill` does not correctly propagate weights into the numerator and denominator histograms of `TEfficiency`.

The fix uses the correct API:

```
eff->SetUseWeightedEvents();
eff->FillWeighted(passed, weight, x);
```

The `SetUseWeightedEvents()` call configures `TEfficiency` to use `Sumw2()`-enabled internal histograms, and `FillWeighted` correctly accumulates weighted counts. This fix affects all efficiency calculations involving cross-section-weighted MC samples and is essential for the blended single+double interaction efficiency study where the mixing weights (e.g., 0.187 and 0.813 for the 0 mrad configuration) must be correctly propagated.

8 Conclusions

The investigation into the double-interaction MC efficiency reveals the following:

1. **The efficiency drop is a matching artifact, not a physics inefficiency.** The reconstruction efficiency falls from ∼94% to ∼31% because vertex displacement in double-interaction events shifts the reconstructed cluster η beyond the $\Delta R < 0.1$ truth-matching threshold. 92% of unmatched photons have their signal cluster present and intact in the calorimeter.

2. **Signal clusters exist and are well-reconstructed.** The shower-shape BDT efficiency is identical (81–92%) for matched clusters in both single and double-interaction MC, demonstrating that the calorimeter response and cluster reconstruction are robust against pileup.

3. **True pileup effects are small.** Isolation efficiency degrades by ∼4% (absolute) due to additional energy from the second collision, and the missing-cluster rate is 8% of the unmatched fraction (i.e., ∼5% of all double-interaction truth photons). Both effects are sub-dominant.

4. **Single-interaction MC is appropriate for efficiency corrections.** Since the efficiency drop is an artifact of the truth-matching procedure and not of the detector response, the single-interaction MC correctly captures the true detector efficiency. Real pileup effects (isolation degradation, cluster loss) are small and are treated as systematic uncertainties after proper weighting by the physics double-interaction fractions (18.7% at 0 mrad, 7.2% at 1.5 mrad).